

CARVE

Cost-aware Active Risk-controlled Verification with Evidence Acquisition

A falsification-first study of refund and credit-not-processed disputes under heterogeneous evidence, exact financial invariants, and asymmetric merchant risk

Abstract. We study *Financial Evidence Consistency Learning (FECL)*: deciding whether customer communications, refund records, identifiers, temporal events, orders, and policies describe one internally consistent financial reality. CARVE combines grounded semantic relation induction with immutable financial invariants, Z3 contradiction certificates, residual-risk estimation, selective abstention, and cost-aware evidence acquisition. FECL-Bench v4.5 contains 2,480 synthetic cases with family-isolated TRAIN/DEV/CALIBRATION/TEST/ODD splits and causal minimal pairs. A one-shot frozen TEST falsifies the broad learned-stack hypothesis: literal rules and formal proof achieve F1 1.000, while relational XGBoost drops to 0.748 and the semantic-only transformer to 0.684. Transformer relation induction shows a smaller macro-F1 lift (0.547 vs. 0.484), but learned relation features do not improve the decision. Conformal risk control certifies no safe PASS coverage, so the component is rejected. A frozen targeted acquisition policy resolves all 480 TEST cases with zero false-PASS exposure, though cheapest-first is slightly cheaper, a retained negative generalization result. The deployable conclusion is deliberately narrow: semantic extraction may assist grounding, but deterministic truth and human review retain authority.

Contribution	Empirical disposition
FECL formulation and v4.5 benchmark	Retained; synthetic only
Transformer relation induction	Retained for relation semantics only
Relational XGBoost	Rejected: no lift over literal rules
Z3 minimum contradiction certificate	Retained: 920/920 pre-TEST exact
CRC selective PASS	Rejected: zero certified PASS coverage
Learned acquisition policy	Not run: simpler policy sufficient

Artifact identity: TEST SHA-256 1fcf4a4f0ca483327116cc91327a9fce545a6f04585eb2732e4bd96dbff06885
One-shot receipt: a27cfe99f74118821454fdf9c1ac939ff74ebb9aa62b0982f4fb95fe9e113f71

1. Problem formulation

Chargeback classification asks whether a dispute belongs to a class. FECL asks a different question: whether heterogeneous evidence is jointly satisfiable under authoritative payment state. This distinction matters because a fluent model can classify language while missing an exact one-rupee, currency, parent-payment, or temporal contradiction. Razorpay's public dispute guidance requires evidence for refund-not-processed claims, and its refund documentation distinguishes refund and payment identifiers [1,2]. We therefore model evidence verification, not outcome prediction.

1.1 Decision problem

Let E_t be the evidence visible at step t , $C(E_t)$ grounded claims, $A(E_t)$ authoritative facts, and I immutable financial invariants. The formal state is:

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Phi(E_t) = Claims(C(E_t)) AND Authority(A(E_t)) AND Invariants(I).
BLOCK if Phi(E_t) is UNSAT; PASS only if Phi(E_t) is SAT and a valid risk certificate permits PASS;
otherwise REVIEW.
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No neural, language, tree, or acquisition model can convert an UNSAT result into PASS. Missing, malformed, OOD, hash-invalid, or unsupported evidence routes to REVIEW. BLOCK is a local evidence-integrity hold, not a legal verdict or financial action.

1.2 Cost-sensitive objective

For decisions d in {PASS, REVIEW, BLOCK}, labels y in {consistent, contradiction}, and acquired evidence S , CARVE minimizes the preregistered synthetic proxy:

$$L(d, y, S) = 1[d=PASS, y=contradiction] * \min(\text{value}, \text{INR } 50,000) + 1[d=BLOCK, y=consistent] * \min(\text{INR } 500, 0.05 * \text{value}) + 1[d=REVIEW] * \text{INR } 100 + \sum(e \text{ in } S) \text{ cost}(e).$$

These are benchmark proxies, not observed losses or claimed merchant savings. The primary safety statistic is false-PASS count and synthetic exposure.

2. Technical whitespace and related work

Transformer encoders such as MiniLM compress self-attention knowledge for efficient semantic representations [3]. XGBoost supplies a strong tabular relational learner [4]. Z3 provides satisfiability modulo theories for machine-checkable contradictions [5]. Conformal risk control extends conformal ideas to bounded monotone risks [6]. Active feature acquisition studies sequentially obtaining costly missing features [7]. CARVE does not claim novelty for these components. Its applied systems contribution is their safety-ordered composition and, more importantly, an empirical demonstration that most learned complexity should be rejected on this benchmark.



Hard precedence: no learned component may override an UNSAT financial invariant.

Unlike a generic chargeback responder, CARVE produces an exact span, an authoritative fact set, a compiled invariant, and a deletion-minimized UNSAT core. LLMs are outside the evaluation and may only verbalize an existing certificate.

3. FECL-Bench v4.5

The benchmark is fully synthetic and balanced by construction; it does not estimate production prevalence. Every causal pair stays within one split. Template family, entity prefix, and pair identity are split-isolated. TEST was not read until runner, model, calibration, and artifact hashes were frozen.

Split	Cases	Pair groups	Families	Purpose
TRAIN	1,200	600	5	Fit supervised models
DEV	320	160	3	Architecture and policy selection
CALIBRATION	320	160	2	Calibration and CRC threshold
TEST	480	240	5 unseen	One frozen execution
OOD	160	n/a	8 categories	Fail-closed safety

3.1 Phenomena and evidence

Cases cover amount mismatch, one-rupee boundaries, partial and cumulative refunds, currency, RRN, ARN/UTR, refund identity, parent-payment identity, wrong-order distractors, temporal contradictions, due and overdue promises, stale status, source disagreement, policy eligibility, negation, Hinglish, OCR corruption, inert prompt injection, missing evidence, and unseen relation composition. Evidence acquisition costs range from 1 for authenticated payment/refund state to 25 for a bank statement.

3.2 Integrity corrections

Versions v4 through v4.4 are preserved as invalid drafts. Pre-fit audits found stale state construction, nonminimal required evidence, a label-directed proof dispatch, multi-cause pairs, row identity mismatch, and label-dependent invariant names. v4.5 is the first frozen candidate to pass 9 integrity suites, 1,840 non-TEST label-blind decision checks, and 920/920 exact contradiction-certificate checks. No v4-family model was fitted before these corrections.

3.3 Leakage controls

Forbidden compiler/model input	Reason
ground_truth_label / material_contradiction	Direct target leakage
phenomenon	Would reveal invariant family
hard_constraints / MCC annotations	Would reveal expected proof
required_for_resolution / oracle trajectory	Would reveal acquisition target
TEST/OOD before final receipt	Would permit selection on holdout

4. Method

4.1 Grounded relation induction

Candidate claim spans are exact document substrings. Word/character TF-IDF and a pinned MiniLM encoder with logistic heads predict semantic relation type; neither creates offsets. Numeric amounts, identifiers, currencies, and dates are parsed deterministically. The transformer is promoted only if DEV relation macro-F1 improves by at least 0.02 and exact-span grounding is at least 0.98.

4.2 Formal compiler and MCC

Applicable invariants are selected from grounded claim content, not benchmark labels. Tracked Z3 assertions bind the claim value, bind the authoritative value, and impose the invariant. On UNSAT, deletion minimization removes any assertion whose removal preserves UNSAT. The resulting three-fact Minimum Contradiction Certificate contains the exact source span, authoritative evidence ID, invariant ID, and proof hash.

5. Experimental protocol

The preregistered ladder is: literal rules; TF-IDF/logistic regression; semantic-only MiniLM; deterministic relational XGBoost; MiniLM relation features plus XGBoost; historical ESRAN; formal proof; selective risk control; and acquisition. ESRAN is reported as historical-not-comparable rather than silently adapted. XGBoost uses depth 3, 240 estimators, learning rate 0.04, subsample 0.85, column sample 0.9, and five seeds. False-PASS weight is chosen from {2,4,8} on DEV. Calibration uses only CALIBRATION. TEST selects nothing.

Metrics include precision, recall, F1, PR-AUC, false PASS/BLOCK, synthetic INR exposure, Brier, NLL, ECE, risk-coverage, relation F1, exact grounding, MCC exact match, OOD REVIEW, acquisition count/cost, and counterfactual repair. Statistical tests use exact paired McNemar and 2,000 minimal-pair bootstrap resamples. Intervals describe only this generator.

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Algorithm 1: CARVE inference
1 validate provenance hashes; if invalid return REVIEW
2 induce grounded relations and parse exact financial attributes
3 compile applicable invariants into  $\Phi(E_t)$ 
4 if UNSAT return BLOCK with MCC
5 if unsupported or OOD return REVIEW
6 if SAT and calibrated risk certificate permits return PASS
7 choose minimum-cost missing evidence; reveal it; repeat
8 append every transition to the audit log

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6. Results

6.1 DEV tournament

System	P	R	F1	PR-AUC	FPASS	FBLOCK	INR exposure
literal deterministic rules	1.000	1.000	1.000	1.000	0	0	0
tfidf lr	0.528	1.000	0.691	0.949	0	143	0
semantic only transformer	0.502	1.000	0.668	0.839	0	159	0
deterministic relational xgboost	1.000	1.000	1.000	1.000	0	0	0
learned relation xgboost	1.000	1.000	1.000	1.000	0	0	0
formal proof	1.000	1.000	1.000	1.000	0	0	0
residual risk initial	1.000	0.225	0.367	0.666	124	0	1,766,931

Literal rules, relational XGBoost, learned-relation XGBoost, and formal proof all reach DEV F1 1.000. The 2,000-pair bootstrap difference between rules and XGBoost is 0.000 with CI [0.000, 0.000]. Therefore XGBoost and its learned-relation extension are rejected. Across five XGBoost seeds, F1 standard deviation is 0.000; the hybrid is less stable, with mean F1 0.988 and standard deviation 0.024.

Transformer relation induction improves DEV macro-F1 from 0.719 to 0.907 with exact span grounding 1.000, passing the relation-specific promotion gate. It does not gain financial decision authority.

6.2 Frozen one-shot TEST

System	P	R	F1	PR-AUC	FPASS	FBLOCK	INR exposure
literal deterministic rules	1.000	1.000	1.000	1.000	0	0	0
tfidf lr	0.675	0.925	0.780	0.943	18	107	293,488
semantic only transformer	0.541	0.929	0.684	0.809	17	189	258,489
deterministic relational xgboost	0.597	1.000	0.748	0.978	0	162	0
learned relation xgboost	0.615	1.000	0.762	0.922	0	150	0
formal proof	1.000	1.000	1.000	1.000	0	0	0
residual risk initial	1.000	0.237	0.384	0.673	183	0	2,493,395

The unseen-family TEST separates structural checks from fitted models. Rules and proof remain at F1 1.000 with zero false PASS and zero false BLOCK. Relational XGBoost falls to F1 0.748 through 162 false BLOCKs; the learned-relation hybrid reaches 0.762 with 150 false BLOCKs. Semantic-only MiniLM has 17 false PASS and synthetic exposure INR 258,489. TF-IDF has 18 false PASS and exposure INR 293,488. The paired McNemar comparison favors proof over XGBoost on 162 discordant cases (exact p rounded to 0.0 in the artifact). The pair-bootstrap F1 delta for XGBoost minus rules is -0.252, 95% CI [-0.268, -0.236].

TEST relation macro-F1 is 0.547 for the transformer versus 0.484 for TF-IDF. The direction persists but the absolute performance and margin shrink substantially, supporting only a bounded assistive role.

7. Selective risk and evidence acquisition

7.1 Calibration and abstention

The initial-view residual model is weak: TEST F1 0.384, PR-AUC 0.673, and 183 false PASS at threshold 0.5. Platt calibration yields TEST ECE 0.029, but calibration alone does not create safe ranking. Under the preregistered corrected value-weighted risk target 0.025, CALIBRATION certifies zero PASS coverage. CARVE therefore sets no PASS threshold. On TEST it autonomously BLOCKs 48 formal contradictions and routes 432 cases to REVIEW: 10% coverage, zero false PASS, and no manufactured confidence.

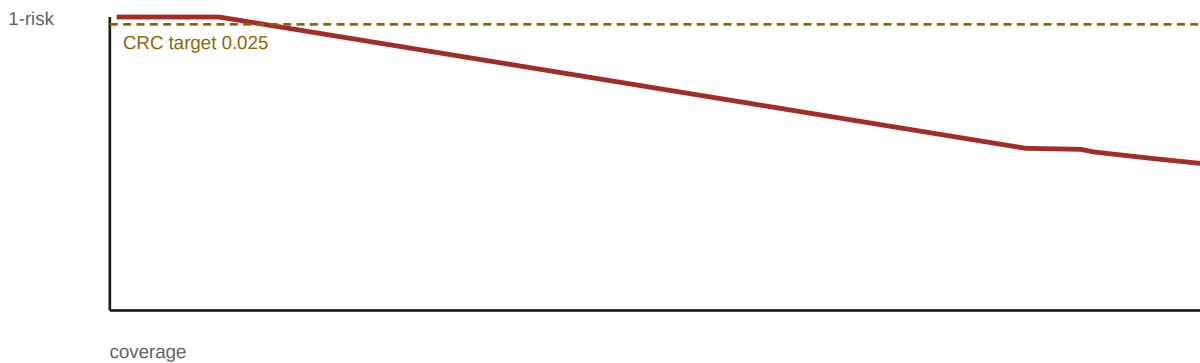


Figure 1. DEV diagnostic risk-coverage curve. It is not a TEST guarantee.

7.2 Sequential acquisition

Policy	TEST cost	Acquisitions/resolved	Resolved	Trajectory exact	FPASS exposure
targeted	996	1.087	480/480	0.662	0
cheapest	924	1.125	480/480	0.550	0
acquire_all	23294	6.354	480/480	0.000	0

DEV selects the targeted non-refund-first policy (cost 456 versus cheapest-first 488). Frozen TEST reverses the ranking: targeted costs 996 and cheapest-first 924. We do not retune after TEST. Both resolve all 480 cases with zero false-PASS exposure; acquire-all costs 23,294. A learned policy is not trained because a simple policy already resolves all cases and the preregistration forbids architecture vanity.

8. Error analysis and ablations

Ablation / failure	Observation	Disposition
Remove formal proof	XGBoost creates 162 false BLOCK on TEST	Reject learned authority
Add learned relations to XGBoost	F1 0.748 -> 0.762; still below rules	Reject hybrid
Semantic classifier only	17 false PASS; INR 258,489 exposure	Research only
Residual risk at 0.5	183 false PASS	Never use raw threshold

Ablation / failure	Observation	Disposition
CRC PASS	0 certified coverage	Reject PASS automation
DEV-selected acquisition	Loses cost ranking on TEST	Retain frozen result
OOD controller	160/160 REVIEW; 0 false PASS	Retain fail-closed boundary

The perfect rule/proof result is not evidence of production perfection. It indicates that v4.5's synthetic contradictions are exactly represented by its explicit invariants. The benchmark is valuable for falsifying unnecessary learned authority but insufficient for claiming general payment-risk performance.

9. Deployment and safety architecture

Production order is provenance validation -> semantic relation extraction -> deterministic verification and proof -> interpretable features -> optional residual risk ranking -> selective controller -> PASS/BLOCK/REVIEW -> evidence acquisition -> append-only audit. Authoritative payment and refund exports are immutable inputs. Every evidence artifact carries a SHA-256 digest; mismatch returns REVIEW. Requests are idempotent and no component can initiate a refund, debit, acceptance, submission, or other financial write.

9.1 Analyst interaction

The product begins with an incomplete case. CARVE names the minimum missing evidence and its acquisition cost. After acquisition, the same compiler either proves an exact contradiction and shows its MCC or returns SAT. Findings link to exact source offsets and authoritative facts. Counterfactual repair changes one causal fact and recomputes the decision; it never rewrites history or performs an autonomous financial action.

9.2 Threat model

Threat	Response
Prompt injection in evidence	Treat as inert text; only grounded claims compile
Model outage	REVIEW; no fallback PASS
Malformed amount / OCR	Reject or REVIEW at parser boundary
Missing ledger state	REVIEW and acquire authoritative state
Hash mismatch	REVIEW before any semantic or financial check
OOD language / schema	REVIEW; 160/160 constructed OOD rejected
Hostile learned score	Cannot override UNSAT or failed invariant

10. Limitations

All counts and currency values are synthetic. Balanced class prevalence, explicit templates, clean structured payloads, and known evidence costs limit external validity. The relation taxonomy is narrow. The CRC calculation is a preregistered empirical controller, not a guarantee under distribution shift; indeed it certifies no PASS. OOD categories are constructed rather than sampled from deployment. We do not evaluate multilingual Indian language coverage beyond synthetic Hinglish, bank network latency, access-control failures, or adversarial corruption of authoritative systems. No claim of novelty, merchant savings, production readiness, or dispute-win improvement follows from this study.

11. Conclusion

CARVE's strongest result is a rejection. Learned relations help interpret language, but modern classifiers and relational boosting do not beat a strong deterministic baseline for exact synthetic financial consistency. Risk control refuses to certify PASS. The surviving system is smaller: grounded semantic assistance, immutable financial invariants, formal contradiction certificates, fail-closed REVIEW, and a simple evidence policy. That is a more credible fintech AI contribution than a larger model stack.

References

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- [4] T. Chen and C. Guestrin. XGBoost: A Scalable Tree Boosting System. KDD 2016, pp. 785-794. DOI 10.1145/2939672.2939785.
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- [6] A. N. Angelopoulos, S. Bates, A. Fisch, L. Lei, and T. Schuster. Conformal Risk Control. arXiv:2208.02814, 2022. <https://arxiv.org/abs/2208.02814>
- [7] O. B. Guney et al. Active Feature Acquisition via Explainability-Driven Ranking. ICML 2025, PMLR 267:20748-20765. <https://proceedings.mlr.press/v267/guney25a.html>

Appendix A. Reproducibility record

Artifact	SHA-256
calibration	0c0f07c7892b5f4a501ded47ef5065326a7ec856002928f89b374633b805db7f
dev	b1e5c8251747a08657e0a177da35b2eff35fcc4d794e56f9c7f404ecd96fea0f
ood	ae14a7f73c3968d9ce8627127868596a2a46438c50ac044f1f04eabfb7825be1
test	d53c5aa914b91035cc153b3587843f97f1b1e08f8f3560228bacb3607d764710
train	300ad749604fa9a0685c3163e17d44034183289eb93a23dfa9748be87aba1892
frozen TEST result	1fc4a4f0ca483327116cc91327a9fce545a6f04585eb2732e4bd96dbff06885
frozen TEST receipt	a27cfe99f74118821454fdf9c1ac939ff74ebb9aa62b0982f4fb95fe9e113f71
Reproduction commands: <pre>uv sync --extra research --extra dev python scripts/generate_fecL_v4_2.py # preserved generator for final v4.5 python -m pytest backend/tests/test_fecL_v4_2_integrity.py backend/tests/test_carve_proof.py -q python scripts/run_carve_v4.py dev python scripts/run_carve_v4.py test --confirm-frozen-test YES # receipt prevents rerun</pre>	

Receipt status: EXECUTED_ONCE. The final TEST command cannot be re-executed without deleting the receipt, which would be an explicit integrity violation.

Appendix B. Reproducibility checklist

Item	Status
Hypotheses and promotion gates preregistered	Yes
TRAIN/DEV/CALIBRATION/TEST/OOD family isolation	Yes
Causal pairs kept within splits	Yes
Generator seed and five stochastic seeds recorded	Yes
Model and feature schema serialized	Yes
Protocol, data, code, models, calibration frozen before TEST	Yes
Single TEST receipt	Yes
Negative and rejected methods reported	Yes
Synthetic-data and no-savings boundary	Yes
External production validation	No